Topological Set Theoretic Estimation:
Follow-on Work of the Past Decade (2015–2026)

Project `thejeb`

July 28, 2026

Abstract

A targeted survey of work in the decade since Carlson's 2012 topological-space recast of set theoretic estimation (STE) [3]. Two findings dominate. First, the *topological*, non-convex branch of STE is, as far as we can determine, an almost single-author thread: Carlson and his direct collaborators, with little independent uptake. Second, no proof assistant has mechanized the multi-set feasibility machinery that STE sits on, which makes the present project's Lean development first-of-kind. Claims below are attributed to primary sources; nothing here is asserted proven in our library unless it is machine-checked in `Ste`.

1 The topological-STE program (2014–2026)

The dissertation grew into a sustained line of follow-on papers with a consistent thesis: reduce every cipher to an equivalent *substitution* cipher, then attack it with STE, the AEP, and unicity distance.

- Carlson et al. [4] — journal version of the dissertation's known-ciphertext attack.
- Ghosh et al. [7] — product ciphers reduce to a single substitution cipher via a "meta-alphabet" construction. This is the published counterpart of what we mechanize in `Ste.Carlson.Reduction` (Theorem 5.2, Corollary 5.3).
- Carlson et al. [1] — uses the isomorph count (the content of Carlson's Lemma 4.1, our `card_consistent_keys`) to shrink the effective brute-force key space.
- Carlson et al. [5] — the most substantive information-theoretic follow-on: Shannon's global unicity distance is an *average*, not a lower bound, and a per-string *local* unicity is defined. A natural (deep) future mechanization target.
- Carlson et al. [6] — security implications of the product-to-substitution equivalence.
- The mode-breaking series, culminating in Hiromoto et al. [8], "Breaking the Counter (CTR) Mode."
- Carlson et al. [2] — a technical report that restates STE (citing Combettes 1993 and Witsenhausen–Schweppe 1968), ties STE explicitly to the AEP and unicity, and frames STE as "a precursor to the neural network"; it dates the *topological*-space STE idea to **2006**, predating the dissertation.

2 Independent uptake of topological STE

We found **no evidence** of anyone outside Carlson's immediate collaboration developing STE over topological (non-metric, non-convex) spaces in this decade. The classical Hilbert-space, convex-feasibility STE lineage (Combettes 1993; see our companion survey) continues to be cited widely, but those citations remain within the convex-optimization tradition; none pivot to the topological generalization. We record this as a genuine gap and opportunity rather than padding it with tangential citations. (Absence of evidence from targeted search is not proof of absence, but multiple queries returned nothing.)

3 STE, the AEP, and unicity after 2012

Post-2012 work tying STE to information theory — property sets as AEP typical sets, and unicity as feasibility collapse — is substantial but lives entirely inside the Carlson cluster, most explicitly in Carlson et al. [5] and Carlson et al. [2]. We found no independent work connecting STE specifically to the AEP, unicity, or decryption in this window.

4 Adjacent topological / feasibility work

Being deliberately conservative: there is essentially no adjacent literature that is genuinely "STE in spirit, topological in substance" outside Carlson's output. Keyword-adjacent hits (e.g. constructions using connected components of a topological space for encryption) are cipher *designs*, not estimation or feasibility methods, and we do not claim an STE connection for them. We deliberately exclude persistent-homology / topological data analysis, a distinct mathematical tradition with no feasibility link to STE.

5 State of mechanization — the opening

This is the decisive input for the project's direction.

- **Lean / Mathlib.** The single-set *Hilbert projection theorem* is formalized (`Mathlib.Analysis.InnerProductProjection`): the existence and uniqueness of the nearest point in a nonempty complete convex set. This is the base case of every projection method — but *no* iterative multi-set feasibility algorithm (alternating projections / POCS, Bregman cyclic projections, Douglas–Rachford) is in Mathlib.
- Li et al. [9] (Optlib) formalizes convex analysis, subgradients, proximal operators, and first-order convergence *rates* on Hilbert spaces; Verified Optimization contributors [10] (CvxLean) does convex problem *modeling*. Neither is feasibility- or projection-framed.
- **Isabelle/HOL** (AFP) has strong Hilbert-space operator theory built for quantum verification, but nothing on projection-method convergence. **Coq/Rocq** has convex polyhedra via the simplex method (finite-dimensional LP only).

Net: no proof assistant has POCS, Douglas–Rachford, Bauschke–Combettes monotone-operator feasibility, or — a fortiori — topological STE. The single-set projection theorem is exactly the base case we would build on. Our roadmap (POCS convergence over `Ste.Convex`, then the topological generalization) is therefore unclaimed ground.

Method note

This survey is exploratory and, where negative, honest about it: several of the conclusions above are *absence-of-evidence* findings from targeted search, flagged as such. As always, no mathematical claim is treated as established for this project until it is machine-checked in the `Ste` library.

References

[1] Albert Carlson, Bhaskar Ghosh, Indira Kalyan Dutta, Shivam Khare, and Michael Totaro. Key space reduction using isomorphs. In *IEEE Int. Conf. on Information Technology, Electronics and Mobile Communication (IEMCON)*, 2021.

[2] Albert Carlson, Ryan Carlson, and Hans C. Mumm. A survey of entropy failure as an attack surface: Artificial intelligence and modern cryptanalysis. Technical report, Center for Governance of Technology and Systems (GoTech), University of Maryland, 2026. Institutional technical report; `https://gotech.umd.edu/sites/default/files/2026-05/EntrokeyAI%20(8).pdf`.

[3] Albert H. Carlson. *Set Theoretic Estimation Applied to the Information Content of Ciphers and Decryption*. Ph.d. dissertation, University of Idaho, Moscow, Idaho, USA, May 2012. Major professors R. E. Hiromoto and R. B. Wells. Retrieved via Internet Archive snapshot (2024-07-13) of the author's ResearchGate full text, publication 349064680.

[4] Albert H. Carlson, Robert E. Hiromoto, and Richard B. Wells. Breaking block and product ciphers applied across byte boundaries. *International Journal of Computing*, 12(3):259–266, 2014. Journal version of the IEEE IDAACS 2011 paper.

[5] Albert H. Carlson, Sarangi R. Mikkilineni, Michael W. Totaro, Robert E. Hiromoto, and Richard B. Wells. An introduction to local entropy and local unicity. In *IEEE Int. Symp. on Networks, Computers and Communications (ISNCC)*, 2022. doi: 10.1109/ISNCC55209.2022.9851755.

[6] Albert H. Carlson, Sarangi R. Mikkilineni, Michael W. Totaro, Richard B. Wells, and Robert E. Hiromoto. Equivalence of product ciphers to substitution ciphers and their security implications. In *IEEE Int. Symp. on Networks, Computers and Communications (ISNCC)*, 2022. doi: 10.1109/ISNCC55209.2022.9851719.

[7] Bhaskar Ghosh, Indira Kalyan Dutta, Shivam Khare, Albert Carlson, and Michael Totaro. Isomorphic cipher reduction. In *IEEE Int. Conf. on Information Technology, Electronics and Mobile Communication (IEMCON)*, pages 947–953, 2021. doi: 10.1109/IEMCON53756.2021.9623135.

[8] Robert E. Hiromoto, Albert H. Carlson, and Manish Singh. Breaking the counter (CTR) mode. In *IEEE Computing and Communication Workshop and Conference (CCWC)*, 2025. doi: 10.1109/CCWC62904.2025.10903932.

[9] Chenyi Li et al. Formalization of convergence rates of four first-order algorithms for convex optimization (optlib). *arXiv preprint arXiv:2403.11437*, 2024. Lean 4; `https://github.com/optsuite/optlib`.

[10] Verified Optimization contributors. CvxLean: Convex optimization modeling in Lean 4. `https://github.com/verified-optimization/CvxLean`, 2023.